

Thomas Karl Atkins

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EDUCATION

University of California-Berkeley

Fall 2025 -

Ph.D. in Computational Biology

Princeton University

Fall 2023 - Spring 2025

M.A. in Quantitative and Computational Biology

University of Minnesota - Twin Cities

Fall 2019 - Spring 2023

Bachelor of Science in Computer Science

Bachelor of Science in Statistical Science

Computational Biology Minor

Thesis: FIST- n D: A tool for n -dimensional spatial transcriptomics data imputation via graph-regularized tensor completion. (<https://hdl.handle.net/11299/254602>)

RESEARCH EXPERIENCE

with Drs. Julien Ayroles and Lydia Lynch

Department of Integrative Biology, UC-Berkeley

Lewis-Sigler Institute for Integrative Genomics, Princeton University

Jan 2024 -

How do we understand variation in immunological systems?

- Working with data from communities in Northern Kenya who are undergoing transitions from pastoralist to market-integrated lifestyles
- Integrating genomic, transcriptomic, and immunological data to determine the impact of genetic and environmental factors on immunological variation

with Dr. Marc Riedel

May 2023 - Aug 2023

Department of Electrical Engineering, University of Minnesota

Do HLA-peptide binding predictors perform better on alleles common in European populations?

- Found strong geographic biases in frequency of training data used to train popular HLA-peptide binding prediction model NetMHCpan and NetMHCIIpan
- Evaluated performance of these models on non-European alleles, finding no difference in prediction accuracy between populations
- Published results in *Frontiers in Immunology* as first and corresponding author

with Dr. Rui Kuang

August 2021 - April 2023

Department of Computer Science, University of Minnesota

Does imputing 3D spatial transcriptomic data give us accurate and useful results?

- Extended a tensor decomposition and graph regularization based algorithm for spatial transcriptomic data imputation from two to three spatial dimensions, making it the first method of its kind
- Implemented this algorithm in an easy-to-use, documented command line tool for 2D and 3D ST data imputation
- Measured performance of our method as superior to existing scRNA-seq imputation methods
- Was awarded an NSF Research Experience for Undergraduates (REU) supplement

with Dr. Julie Segre

June 2022 - Aug. 2022

National Human Genome Research Institute, National Institutes of Health

How do we diagnose errors in fungal genome variant calling pipelines?

- Developed an extensible all-in-one variant calling pipeline for population genomics by filtering different combinations of bcftools, freebayes, and GATK variant callers
- Created a suite of diagnostic plots that identified inconsistent results in popular workflows and assumptions from human genetics that did not hold in a fungal context

PUBLICATIONS & PREPRINTS

- Proctor DM, Sansom SE, Deming C, Conlan S, Blaustein RA, **Atkins TK**, ... Hayden MK, Segre JA. Clonal *Candida auris* and ESKAPE pathogens on the skin of residents of nursing homes. *Nature*, 26 Feb 2025. DOI: 10.1038/s41586-025-08608-9.
- **Atkins TK**, Solanki A, Vasmatzis G, Cornette J, Riedel M. Evaluating NetMHCpan performance on non-European HLA alleles not present in training data. *Frontiers in Immunology*, 15 Jan 2024. DOI: 10.3389/fimmu.2023.1288105.
- **Atkins TK**, Song T, Kuang R. FIST-*n*D: A tool for *n*-dimensional spatial transcriptomics data imputation via graph-regularized tensor completion. *bioRxiv*; 16 Oct 2022. DOI: 10.1101/2022.10.12.511928.

PRESENTATIONS

- **Atkins TK**, Garske KM, Peng JZ, Mwai C, Chao M, Gerlinger E, Kahumbu J, Mukoma B, John E, Kinyua P, Lopurudoi A, Mutai N, Martins D, Lea A, Ayroles JF. Dysregulation of cellular signaling networks upon rapid environmental shift. Presented at CHSL Biology of Genomes; 07 May 2025.
- **Atkins TK**, Garske KM, Peng JZ, Mwai C, Chao M, Gerlinger E, Kahumbu J, Mukoma B, John E, Kinyua P, Lopurudoi A, Mutai N, Martins D, Lea A, Ayroles JF. Environmental change rewires Turkana CD4⁺ gene regulatory networks. Presented at GRC Quantitative Genetics; 15 Feb 2025.
- **Atkins TK**, Garske KM, Peng JZ, Chao M, Kahumbu J, Lea A, Ayroles JF. A multiomic atlas of a Northern Kenyan cohort. Presented at Chan-Zuckerburg Initiative Cell Science Meeting; 10 Sep 2024.

- **Atkins TK**, Song T, Kuang R. FIST- n D: A tool for n -dimensional spatial transcriptomics data imputation via graph-regularized tensor completion. Presented at University of Minnesota Undergraduate Research Symposium; 9 Dec 2022; Virtual. Url: <https://ugresearch.umn.edu/presentation-opportunities/fall-symposium/presenters-2022/thomas-atkins>
- Proctor DM, **Atkins TK**, Samson SE, Hayden MK, Segre JA. Integrating Data Types to Understand the Genomic Epidemiology of the Emerging Fungal Pathogen *Candida auris*. Presented at: CSHL Microbiome; 25-29 Oct 2022.
- **Atkins TK**, Proctor DM, Deming C, Chen Q, Conlan SP, Segre JA. Diagnostic Measures for Fungal Genome Variant Callers. Presented at: NIH Summer Poster Day; 3 Aug 2022; Virtual.
- **Atkins TK**, Chan-Santiago JA, Megret R. Re-Identification of Honeybees via Attention-Based Feature Aggregation along Trajectories. Presented at: IQ BIO REU, NEURO-ID, & IBD2K Research Presentations; 29 Jul 2021; Virtual.

TEACHING EXPERIENCE

CSCI 2021: Machine Architecture and Organization

Spring 2021

Teaching Assistant under Dr. Christopher Kauffman

- Co-led lab sections on low-level programming concepts
- Offered feedback on course design and problem descriptions
- Assisted students at office hours

CSCI 1933: Introduction to Data Structures and Algorithms

Fall 2020

Teaching Assistant under Dr. Chris Dovelis

- Gave weekly lab presentations
- Designed and implemented a lab manual covering an introduction to Object-Oriented Programming
- Created a course feedback survey to better understand student needs and satisfaction
- Met with students three times over the semester to discuss course performance and feedback
- Assisted students at office hours

COURSEWORK

Princeton University

- Introduction to Genomics and Computation, Method and Logic in Computational Biology, Biological Networks across Scales, Research Topics in the Quantitative Life Sciences, Foundations of Statistical Genomics

University of Minnesota

- **Computational Biology:** Functional Genomics, Systems Biology, and Bioinformatics; Computational Techniques in Genomics; Bioinformatic Analysis; Personal Genome Analysis; Large Scale Omic Data in Plant Biology; Theoretical Neuroscience; Pharmacogenomics; Circuits, Computation, and Biology
- **Computer Science:** Algorithms and Data Structures; Data Visualization; Special Topics in Computer Science (Teaching in Computer Science); Honors Discrete Structures of Computer Science; Program Design and Development; Natural Language Processing
- **Statistics and Mathematics:** Regression and Correlated Data; Introduction to Statistical Computing; Applied Statistics I/II; Theory of Statistics I/II; Applied Linear Algebra; Honors Calculus III/IV; Bayesian Astrostatistics; Consultation and Communication for Statisticians
- **Biology:** Honors Introduction to Biology, Genetics; Immunology